FANUC America Corporation Servo Torch Utility Package Installation and Setup Guide

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Patents

One or more of the following U.S. patents might be related to the FANUC America products described in this manual.

FRA Patent List

4,630,567 4,639,878 4,707,647 4,708,175 4,708,580 4,942,539 4,984,745 5,238,029 5,239,739 5,272,805 5,293,107 5,293,911 5,331,264 5,367,944 5,373,221 5,421,218 5,434,489 5,644,898 5,670,202 5,696,687 5,737,218 5,823,389 5,853,027 5,887,800 5,941,679 5,959,425 5,987,726 6,059,092 6,064,168 6,070,109 6,086,294 6,122,062 6,147,323 6,204,620 6,243,621 6,253,799 6,285,920 6,313,595 6,325,302 6,345,818 6,356,807 6,360,143 6,378,190 6,385,508 6,425,177 6,477,913 6,490,369 6,518,980 6,540,104 6,541,757 6,560,513 6,569,258 6,612,449 6,703,079 6,705,361 6,726,773 6,768,078 6,845,295 6,945,483 7,149,606 7,149,606 7,211,978 7,266,422 7,399,363

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VersaBell, ServoBell and SpeedDock Patents Pending.

Conventions

This manual includes information essential to the safety of personnel, equipment, software, and data. This information is indicated by headings and boxes in the text.

Warning

Information appearing under WARNING concerns the protection of personnel. It is boxed and in bold type to set it apart from other text.

Caution

Information appearing under **CAUTION** concerns the protection of equipment, software, and data. It is boxed to set it apart from other text.

Note Information appearing next to **NOTE** concerns related information or useful hints.

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Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system-including the design, implementation, and debugging of application programs-must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and **not** replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.

- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.
- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points* . Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.
- Before teaching, visually inspect the robot and *work envelope* to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

> **Warning**
>
> **Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.**

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

> **Warning**
>
> **Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.**

— Using a low motion speed, single step the program for at least one full cycle.

— Using a low motion speed, test run the program continuously for at least one full cycle.

— Using the programmed speed, test run the program continuously for at least one full cycle.

- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.
- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that an applicator bell cup can continue to spin at a very high speed even if the robot is idle. Use protective gloves or disable bearing air and turbine air before servicing these items.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Warning

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.

- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

Warning

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

Note Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

Chapter 1

Overview

Contents

Servo Torch is a fully integrated wrist-mounted feeder that operates within the control architecture of a FANUC America Corporation controller. Drive roll designs support both steel and aluminum wires (Steel “V” design , Aluminum “U” design). Multiple wire diameters are supported (.035” through .062”). FANUC America Corporation and Lincoln have worked closely to develop advanced Aluminum operating modes. This joint development has provided advanced previously available for aluminum applications. Proper implementation and use of the tools provided is critical.

Caution

Servo Torch is a wire drive designed to feed soft wire. It does not solve nor is it responsible for all Aluminum application problems. Proper implementation and power source selection, combined with advanced aluminum process training is critical for application success.

Required Equipment

Contents

The following parts are typical for a 100*i*B Servo Torch utility package.

Table 2–1. Parts

ITEM	FRA Part Number	Description	Number Required
1	A05B-1215-J301	Servo Torch Mechanical Drive	1
2	A05B-1215-J322	Bracket Assembly (Standard FRA Configuration)	1
3	A05B-2485-J001	AUX.AXIS Amp Assembly	1
4	A06B-6114-H103	SVM1-20I	1
5	A250-8003-X526	SVU Adapter Plate	1
6	A05B-2485-J470	Cable Tag	1
7	A05B-2485-J474	Cable Tag	1
8	A20B-8100-0762	Aux. Axis PCB 4	1
9	A05b-2485-J460	Battery Box	1
10	A05B-1215-J334	14 Meter Servo Torch Cables	1
11	FO-1800-012	Axis 1 cable support bracket	1
12	FO-4813-001	Servo Cable Support Bracket	1
13	TORMO000000001O	BINZEL #783.9600 SVT ADAPTER	1
14	TORMO000000002O	BINZEL #783.9601 100iB SVT UTIL CBLE	1
15	TORMO000000003O	BINZEL #156.9099 SVT CONDUIT	1
16	TORMO000000005O	BINZEL #783.9602 SVT UTILITY BOX	1
17	TORMO000000009O	BINZEL #128.9005 SVT PA LINER -12 foot 3/64" to 1/16" wire	1
18			
19	HDWMO000003036O	Lashing Tie Clamp	1
20	TORMO000000036O	BINZEL #785.5102 SVT VTS500 – 22 deg. Water Cooled	1
21	TORMO000000038O	BINZEL #145.0553 SVT NOZZEL – Coarse thread 1mm recess	1
22	TORMO000000037O	BINZEL #783.9615 SVT VTS LINER – extended tail	1
23	TORMO000000010O	BINZEL #142.0117 TIP HOLDER	1
26			
28	A6-SW2NA-5X10S	Ground Screw	1
29	A6-SW2NA-5X12S	Amp Screw	6

Table 2–1. Parts (Cont'd)

ITEM	FRA Part Number	Description	Number Required
30	A99L-0035-0001	Clamp	1
31	FO-4813-003	In-feed support bracket	1

Chapter 3

References

Contents

Refer to the *SYSTEM R-J3iB Controller Servo Torch Mechanical Unit (Option) Maintenance Manual*.

Chapter 4

Retrofitting the Servo Torch

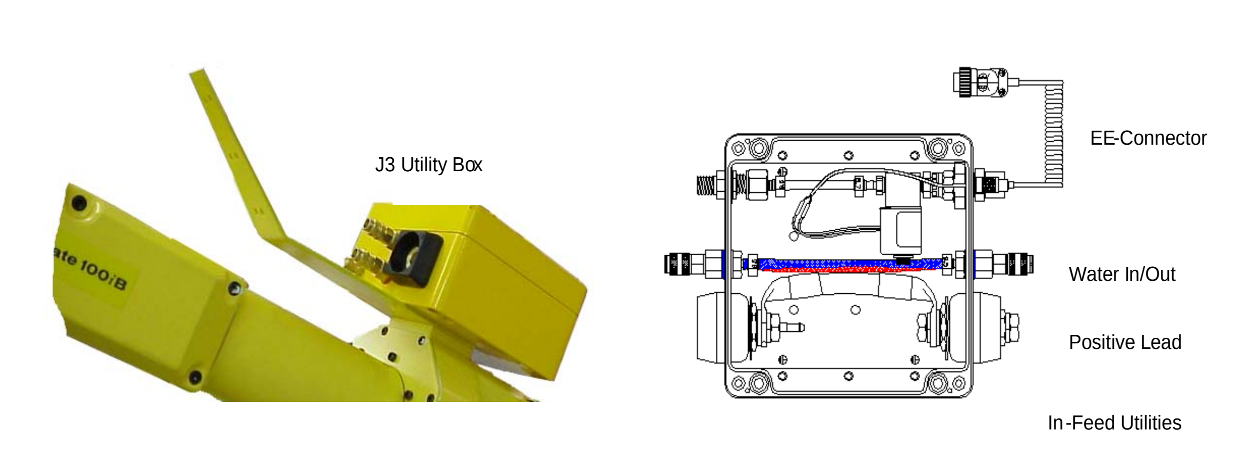
Contents

Procedure 4-1 Mounting Utility Box and Cable Support

The utility box provides a junction separating the dedicated routing from the power supply from the flexible routing to the Servo Drive. It is a straight pass through for weld power, cooling water and contains a 24v solenoid for shielding gas. The gas solenoid is triggered from RO-1 (robot output) and is connected to the robot through the 5 pin, EE connector. Provisions for weld utilities are mounted on the in-feed side of the box. Welding power is connected through a ¾” bolt connection and is insulated with a boot. A female “Western Brand” inert fitting is for shielding gas. And two Rectus male post with clamps are supplied to connect the cooling water lines. Each utility box is shipped with the appropriate in-feed and mounting hardware typically all in a sealed bag.

1. Utilize 4-M6 bolts and sandwich the cable support FO-4813-001 between the utility box and the J3 mounting surface.
2. The Cable support bracket should be installed so the support bend extends towards the robot wrist and is as close the J5 pivot as possible. See Figure 4–1 .

Figure 4–1. Utility Box Mounting



3. AM–120iB series robots require a cable support extension, install the extension as in the photo below.

Procedure 4-2 Mounting the Servo Torch Mechanical Drive on the Wrist of the Robot

Components

1. The bracket assembly kit A05B-1215-J322 is used to attach the drive to the robot.
   a. Install the 1st location dowel in the wrist of the robot.
   b. Install the spacer puck (A290-7215-X741) to the wrist of the robot.
   c. Install the 2nd location dowel in the spacer puck.
   d. Bolt the adapter plate (A290-7215-X715) to the spacer puck (A290-7215-X741).
   e. Loosely bolt one the of the side plates (A290-7215-X717) to the adapter plate.

f. Install the 2 dowels into the adapter plate and tighten bolts to 15.7 Nm. (Loctite 242)

g. Loosely bolt the servo drive (A05B-1215-J301) to the same side plate.

h. Install the 2 dowels into the servo drive and tighten bolts to 15.7 Nm. (Loctite 242)

i. Route the cables (A05B-1215-J334) around the drive motor and out the access.

j. Loosely bolt on the 2nd of the side plates (A290-7215-X717).

k. Install the remaining 4 dowels and tighten to 15.7 Nm.

l. Bolt motor covers in place (A290-7215-X709) (A290-7215-X719) (A290-7215-X718).

See Figure 4–2 .

Figure 4–2. Mount the Servo Torch Drive

Procedure 4-3 Install the Binzel Torch and Utility Adapters to the Servo Torch Drive

1. Torch Outlet
   a. Attach the insulated adapter #783.9604 to the FANUC gear box with 4 M5x10 bolts.
   b. Attach the bass VTS adapter # 783.9607 to the insulator with 3 – M5x14 bolts.
2. Wire Inlet
   a. Attach the quick connector adapter block #783.9603 to the FANUC gear box with 4 M5x14 bolts.
   b. Screw the brass Rectus conduit quick connect #177.0012 into the insulated adapter.

See Figure 4–3 .

Figure 4–3. Install the Binzel Torch

Procedure 4-4 Assembling the Drive Rolls

1. Each drive roll set contains 4 drive rolls comprising of 2 drive and 2 tension.
   a. Pre-assemble the tension roller assembly before installing into the drive. The larger bushing must go on the bolt head side.
   b. Tension rollers are designed to only be installed the correct way.
   c. The drive roller gets bolted to the drive bushing.
   d. Push the drive bushing over the drive pin and secure with retaining nut.

   See Figure 4–4 .

Figure 4–4. Assembling the Drive Rolls

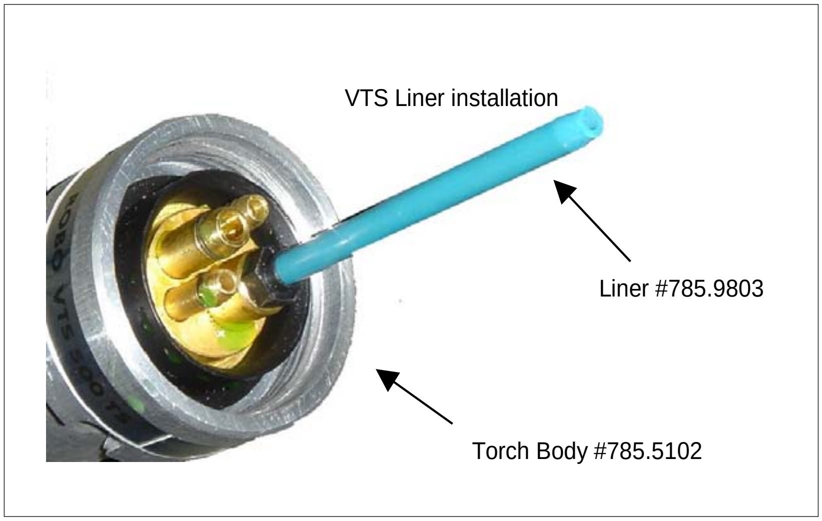
e. With the drive rolls installed confirm that the drive groves are in line with each other. Initial installation of the drive bushing can be tight and sometimes requires additional pressure to assure it has been properly seated over the drive post of the feeder.

f. Failure to install the drive rolls properly can result in deformation of the wire and poor feeding.

Procedure 4-5 Install Aluminum Neck Liner — Binzel VTS500 torch

1. Install the VTS liner #785.9803 into the VTS500 torch body #785.5102.
2. Attach the VTS500 torch to the VTS adapter #783.9607. See Figure 4–5 .

Figure 4–5. Install Aluminum Neck Liner



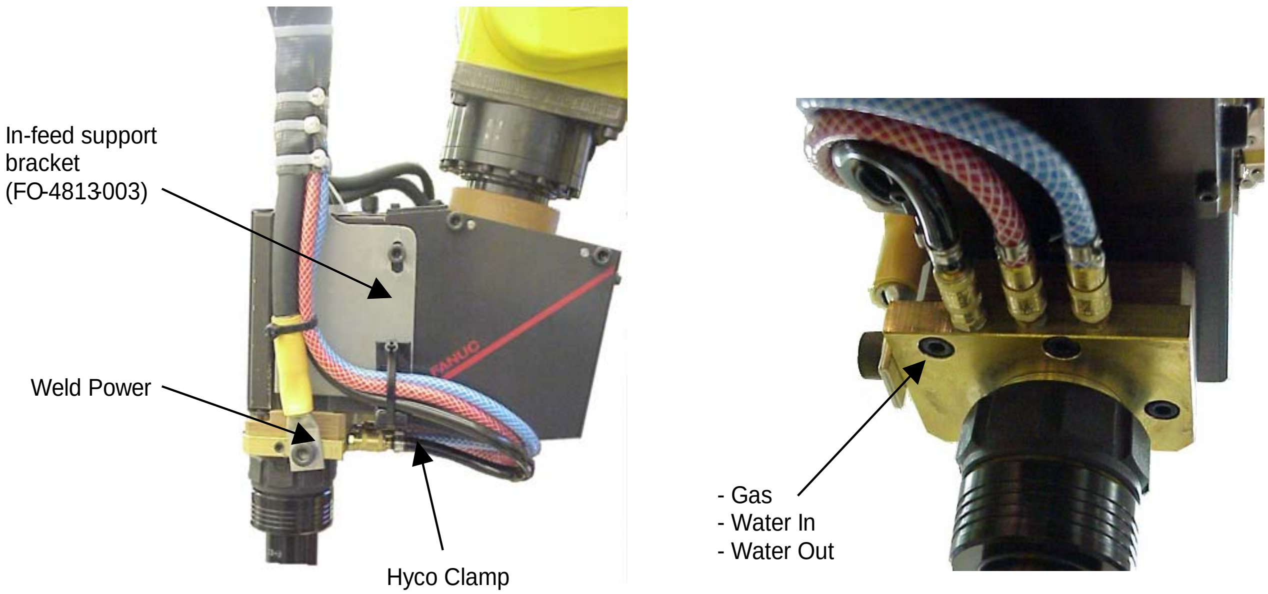
Procedure 4-6 Installing the torch utility cable #783.9601(100iB) or #783.9610 (120iB) and secure external cable routing

Note The purpose of the weld utility cable is to supply welding power, cooling water and shielding gas to the Servo Drive assembly. All connections on this cable are quick connects for ease of assembly and replacement. The weld utility cable runs between the J3 interface and the wrist mounted Servo Drive. When installing the utility cable start at the Servo Drive and work back toward the J3 utility box. Manage any excess cable at the J3 Utility box.

1. Servo Drive Connections
   a. Install the In-feed support bracket (FO-4813-003).
   b. Connect the weld power to the brass VTS adapter #783.9607.
   c. Route the gas and water lines along the side of the gear box and connect to the appropriate quick connectors.
   d. Secure the water and gas lines to the side of the Servo drive with a HEYCO cable clamp #3327.
   e. Secure the Weld Utility bundle with 3 nylon straps to the In-feed support bracket.

See Figure 4–6 .

Figure 4–6. Weld Utility Cable Routing



2. Weld Utility Cable Routing

a. Group the weld utility and the motor cables and route back towards the J3 utility box.

b. Provide a service loop to allow full range of J5 and J6 motion.

c. Secure the cable bundle to the support bracket (FO-4813-001), secure in place with nylon straps.

d. Coil any excess cable before connecting to the J3 utility box.

3. Utility Box Connections

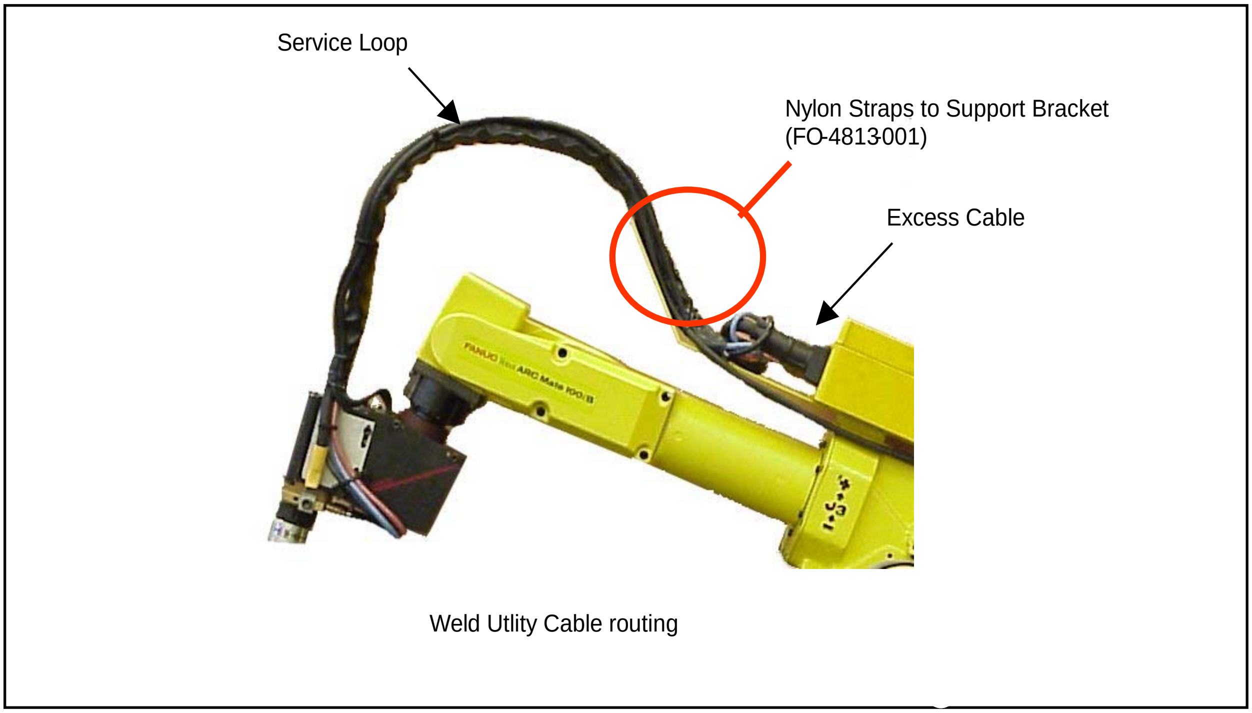
a. Connect the weld power Dins quick connector to the axis 3 utility box.

b. Connect the supply (blue) and return (red) water leads to the utility box.

c. Connect the black gas supply to the utility box.

See Figure 4–7 .

Figure 4–7. Utility Box Connections



Note In-Feed weld utilities are supplied by the integrator and are not part or the servo torch package

4. Routing from J3 to J1

a. Connect the weld power to the ¾” power lug on the J3 utility box.

b. Connect the remote electrode sense lead to this same lug.

c. Connect water supply (blue) and return (red) to the utility box.

d. Connect gas supply to the utility box.

e. Combine the cables provide a motion loop for J3 and secure to the J2 upper arm.

f. Secure a second time lower down on the J2 arm.

g. Install the J1 tail bracket.

h. Provide a J2 motion loop and secure to the tail bracket with nylon straps.

See Figure 4–8 .

Figure 4–8. **Infeed Weld Utilities**

Procedure 4-7 Installing the wire delivery conduit and securing for smooth wire transfer

1. Conduit Assembly

a. **-** Remove the pre-assembled conduit male adapter, this is a 3 piece assembly consisting of the nylon stop #131.9003, brass retainer #881.1096 and the male conduit adapter #881.9003.

b. Using a sharp razor knife cut one end of the SVT PA liner #128.9005, make sure that the cut is clean and that the liner does not get crushed. (This liner is 12 foot long and may have hardware on one end, by cutting the end the hardware is removed)

c. Jam the nylon conduit stop #131.9003 onto the SVT PA liner #128.9005, make sure that the liner bottoms out on the stop

d. Re-assembly the 3 piece assembly, the SVT PA liner should now be part of this assembly

e. Slide the SVT PA liner #128.9005 into the protective SVT Conduit #156.9099, since the liner is 12 foot and the conduit is 8 foot there should be 3 feet plus of PA liner sticking out of the SVT Conduit to work with

f. Pass the SVT PA #128.9005 liner through the second male conduit adapter #881.9003 and thread the adapter onto the SVT Conduit #156.9099

g. With sufficient SVT PA liner sticking out past the threads of the male conduit adapter determine the amount that would be required to press the second Nylon conduit stop on and assure a tight fit once the brass retainer #881.1096 is tightened.

h. Using a sharp razor knife cut one end of the SVT PA liner #128.9005, make sure that the cut is clean and that the liner does not get crushed.

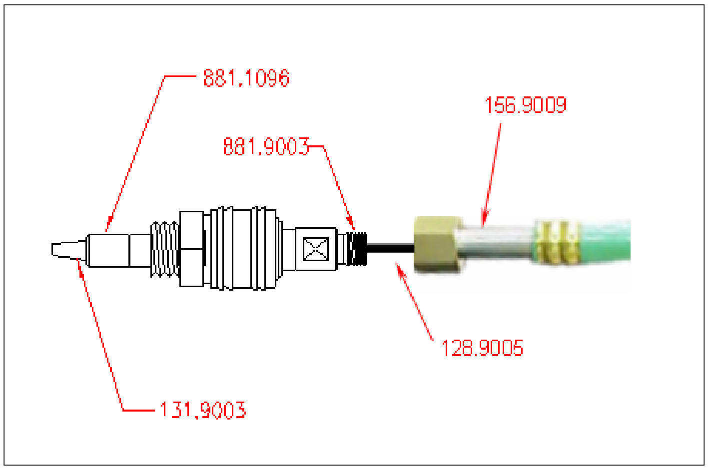
i. Jam the nylon conduit stop onto the SVT PA liner, make sure that the liner bottoms out on the stop.

j. Pass the Nylon stop through the second brass retainer and tighten the brass retainer to the male conduit adapter.

k. The SVT Conduit assembly is complete and can be installed.

See Figure 4–9 .

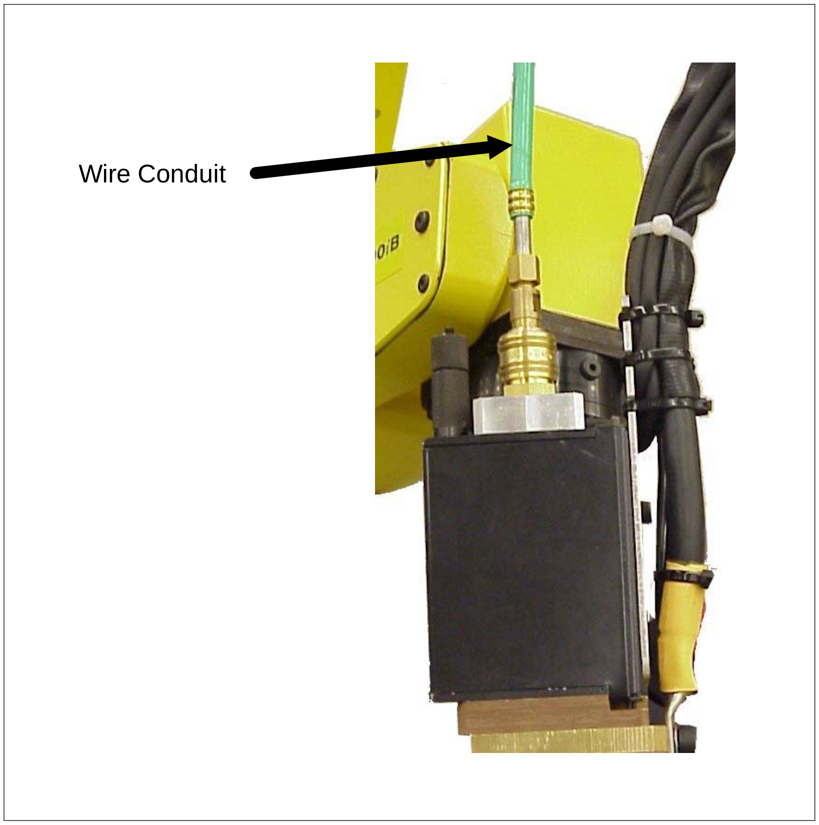
Figure 4–9. Installing the Wire Delivery Conduit



2. Secure the delivery conduit to the Servo Drive

a. Pass the assembled conduit assembly through the RECTUS quick connect #177.0012 to provide a rigid mount to the Servo Torch wire drive. See Figure 4–10 .

Figure 4–10. Securing the Delivery Conduit to the Servo Drive



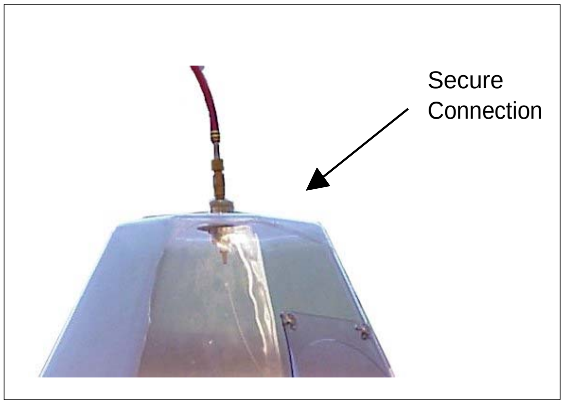
3. Secure the delivery conduit to the wire source.

a. Attach the second RECTUS quick connect #177.0012 to a rigid mount near the wire spool (This mount must be rigid to assure smooth wire transfer)

b. Pass the assembled conduit assembly through the RECTUS quick connect #177.0012 to provide a rigid mount at the wire source

c. Box wire is the preferred source. See Figure 4–11 .

Figure 4–11. Securing the Delivery Conduit to the Wire Source



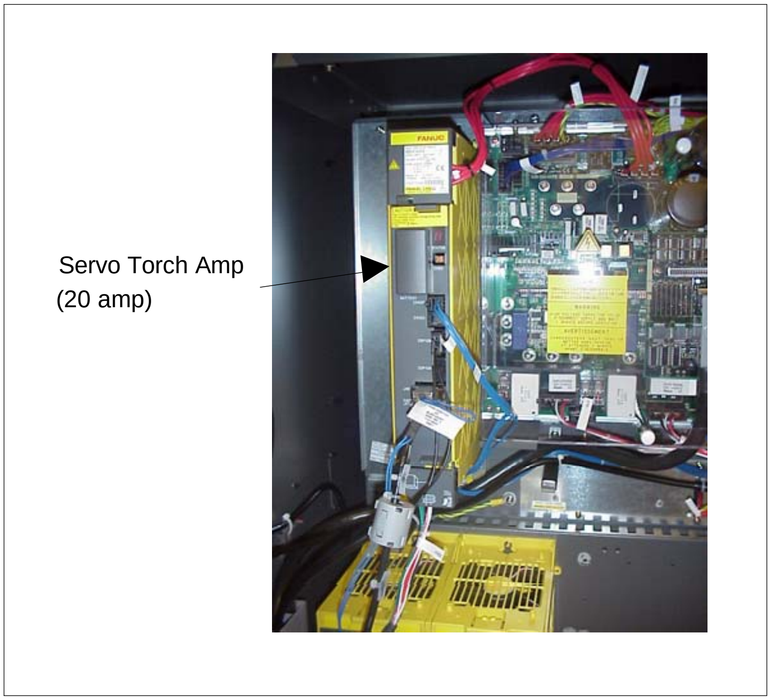
Note Loose connections at either end of the conduit will result in poor wire feeding and may cause excessive arc flaring.

Procedure 4-8 Servo Amp Installation

1. Remove spare amp cover next to the 6 axis amp
2. Replace the cover with A250-8003-X526 SVU adapter plate
3. Mount the amp (SVM1-20I) in the open slot on the adapter plate
4. Install the auxiliary axis wiring A05B-2485-J001
5. Install the 4 axis auxiliary PCB and connect the fiber optic between the PCB and AMP

See Figure 4–12 .

Figure 4–12. Installing the Amp



Procedure 4-9 Establish external voltage feedback lines

1. Electrode (+) feedback must be established since the PF-10 has been replaced with the Servo Torch drive. Failure to provide electrode feedback will result in open circuit voltage when welding.
2. To minimize voltage loss, connect feedback pickup points as close to the welding arc as possible.

   Lincoln PowerWave 455 - Electrode or Positive sense lead can be established by connecting a wire between Pin “C” on the ArcLink connector at the weld power source and the positive lug on the Servo Drive - Work or Negative sense lead is set to internal by default. Refer to the power supply manual on how to establish an external work sense lead

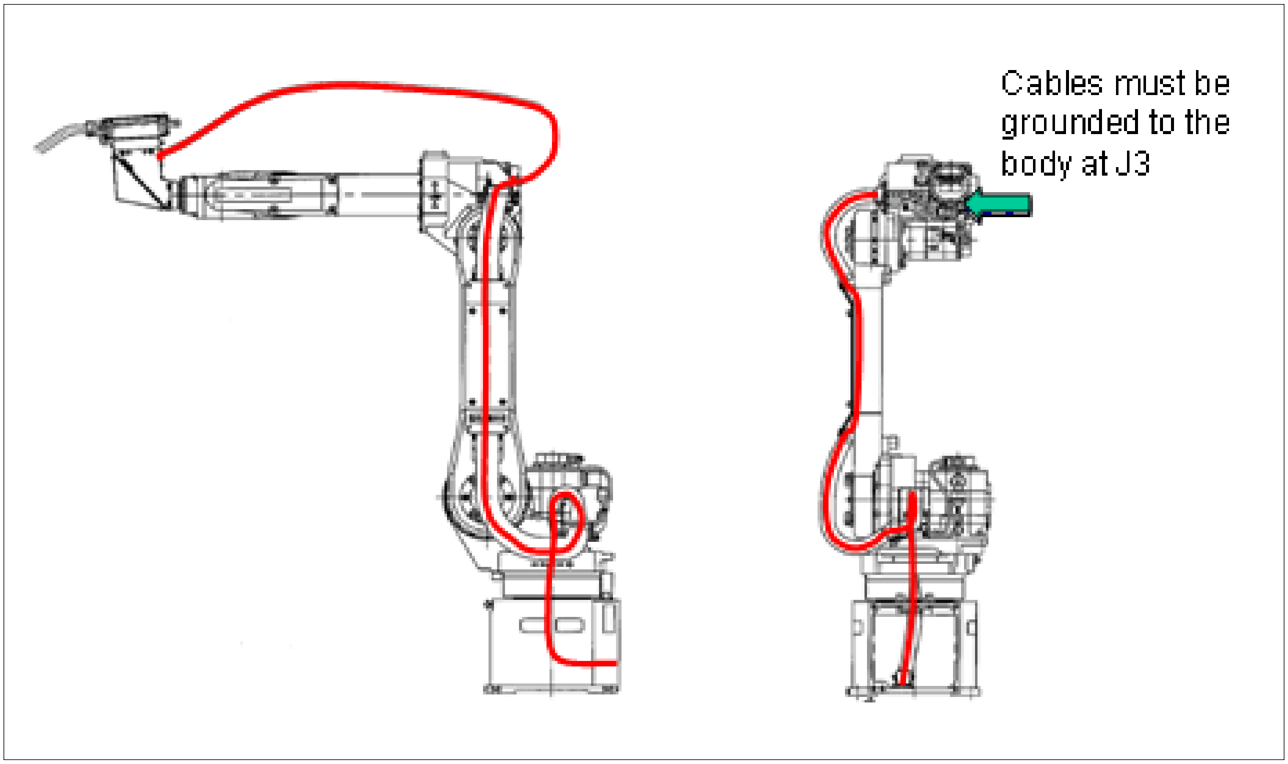
   Note Contact the weld supplier to verify the connection methods are accurate. Failure to properly connect electrode and work sense leads will result in poor power supply performance, and potential damage to welding torches and consumables.

Procedure 4-10 Noise Protection

To prevent weld interference and nuisance noise related alarms proper routing and grounding of the pulse cable is required.

1. **Grounding at the robot** Connect the “Grey” ground wire from the pulse cable to the J3 casing of the robot. See Figure 4–13 .

Figure 4–13. Noise Protection



2. **Grounding at the Control Cabinet**

   a. Ground the Servo Torch amp to the robot control cabinet.

   b. Ground the pulse signal line to the ground strip in the control cabinet.

3. **Routing Techniques**

   a. Isolate the pulse signal cable from power lines as much as possible.

   b. Minimize the parallel distance in which these two cables run together.

   c. Avoid coils.

> **Caution**
>
> Failure to route and isolate signal lines properly might result in nuisance noise related servo alarms.

Chapter 5

Servo Torch Software Installation

Contents

5.1 Installation Overview

The Servo Torch axis is set up at Controlled Start using the Maintenance Menu. During setup you define how many Servo Torch axes are in the system (1 or 2), assign the Servo Torch axis hardware start number, and set the amplifier number. The Servo Torch control functions are set up on the Weld Equipment Setup menu after the axis is configured.

Note Servo Torch axes must be installed after the regular robot axes or extended axes. - In a Dual Arm configuration the Servo Torch axis order is sequential (Table 5–3).

Table 5–1. Example 1 — 6 axis robot + Servo Torch

Hardware Axes	Axis Type
1–6	Robot Axes
9	Servo Torch Axis

Table 5–2. Example 2 — 6 axis robot + Extended Axis (or Nobot or Positioner) + Servo Torch

Hardware Axes	Axis Type
1–6	Robot Axes
9	Extended Axis
10	Servo Torch Axis

Table 5–3. Example 3 — Dual arm 6 axis robots + Servo Torch + 2 Groups

Hardware Axes	Axis Type
1–6	1st Robot Axis
7	Servo Torch Axis 1
8	Servo Torch Axis 2
9–14	2nd Robot Axis
15	Group 3
16	Group 4

Use Procedure 5-1 to set up Servo Torch. Use Procedure 5-2 to set up Torch Guard.

Procedure 5-1 Setting Up Servo Torch

Conditions

- You have ArcTool V6.22–1 or later loaded.
- The Servo Torch Hardware is installed. Refer to the *Servo Torch Mechanical Unit (Option) Maintenance Manual*.

- The Servo Torch software option is installed. Refer to the *Software Installation Manual* for more information.
- Robot power is OFF.

1. Control Start the robot (press and hold PREV and NEXT keys while turning on power. You will see a screen similar to the following.

```
ARCTOOL SETUP CONTROLLED START MENUS
1 F number F00000
  Equipment: 1
2 Manufacturer: General Purpose
3 Model: MIG (Volts, WFS)
Press FCTN then START (COLD) when done
```

2. Press MENU.
3. Select Maintenance. You will see a screen similar to the following.

```
ROBOT MAINTENANCE
Setup Robot System Variables
  Group  Robot Library/Option   Ext Axes
    1    M-6iB/ARCMATE100iB         0
    0    ServoTorch                 1
```

4. From the Robot Maintenance menu, move the cursor to Group 0, Servo Torch.
5. Press F4 (MANUAL).
6. From the Setup Servo Torch Axis menu, respond to the questions as follows:
   - Enter amp number: typically 2, but enter the amplifier number connected to the servo torch motor.
   - Enter hardware start axis: see above table, but normally 9
   - Enter number of axes: 1 or 2
   - Setup type: 1 (Normal Setup)

   You will see a screen similar to the following.

```
- Setup Servo Torch axis —
Select Setup type
0: Exit
1: Normal setup
2: Direct ISDT setup
Setup type? 1
Enter number of axes (1-2)?: 1
Enter hardware start axis ?:7
Enter amp number (axis: 1)?: 2
```

7. Upon completion of setting up the Servo Torch axis, the Teach Pendant display will return to the Robot Maintenance Menu. Press the FCTN key and ENTER to Cold Start the controller.
8. When Cold Start is complete Press MENU, Setup, Type, Weld Equipment. Scroll to the entries for the wire feeder. You will see a screen similar to the following.

```
SETUP Weld Equip
                                   5/20
Feeder: Servo Torch
 5 Wire size:    1.2 mm
 6 Wire material: Aluminum
 7 Wire feed speed units:    IPM
 8 WIRE+ WIRE- speed:       50    IPM
 9 Feed forward/backward:   ENABLED
10 Wire stick reset:        ENABLED
11 Wire stick reset tries:  1
```

9. Move the cursor to line 5, Wire size: Press F4, [CHOICE] and select the filler wire diameter from the options available.
10. Move the cursor to line 6, Wire material: Press F4, [CHOICE] and select the wire material from the options available.
11. Move the cursor to line 12, Servo Torch. Press ENTER to setup the Servo Torch Detail menu. You will see a screen similar to the following.

```
SETUP Weld Equip
                                    12/20
 6 Wire material: Aluminum
 7 Wire feed speed units:   IPM
 8 WIRE+ WIRE- speed:       50    IPM
 9 Feed forward/backward:   ENABLED
10 Wire stick reset:        ENABLED
11 Wire stick reset tries:  1
12 Servo Torch (DISABLED):  <*DETAIL*>
Timing:
```

12. On the Servo Torch Setup page, Press F4, ENABLE to enable the Servo Torch function. You will see a screen similar to the following.

```
SETUP Weld Equip
                                  1/8
Servo Torch setup
  1 ServoTorch function:       ENABLE
  2 ServoTorch axis index:      1
  3 Wire inching mode:         NORMAL
  4       Inch length:         15.000mm
Welding gas signal
  5 Gas start signal:        RO [    1]
Air purge setup
  6 Air purge function:          DISABLE
```

13. Move the cursor to item 2 (Servo Torch axis index) and enter 1.

14. Servo Torch is set to NORMAL wire inching mode by default. This means that wire feeds at the WIRE +/- speed set on the previous menu as long as the wire inch keys are held. CONSTANT is an alternative wire inch mode, where the wire is fed for a constant length as set by the value in item 4, Inch Length. In CONSTANT mode, the wire feeds at the rate set by the WIRE +/- speed value for the length set in item 4 and stops when the length equals this value.

15. Servo Torch controls the shield gas solenoid and by default is wired to Robot Output [1]. You may change this setting if your system is wired differently.

16. Servo Torch provides an optional air purge that is designed to blow out the wire feed drive roll cavity to reduce wire flaking buildup. This feature may be enabled on line 6 of the Servo Torch setup menu.

17. If you choose to enable the Air Purge function, you may assign the output that controls the air valve and the air purge time on lines 7 and 8 of the Servo Torch setup menu. You will see a screen similar to the following.

```
SETUP Weld Equip
                                  5/8
  2 ServoTorch axis index:      1
  3 Wire inching mode:         NORMAL
  4       Inch length:         15.000mm
Welding gas signal
  5 Gas start signal:        RO [    1]
Air purge setup
  6 Air purge function:          DISABLE
  7 Air purge signal:        RO [    2]
  8 Post flow time:              0.50sec
```

18. When you have completed the Servo Torch Detailed Setup, you must cycle power to the robot controller to make the selection active.

19. When power is restored to the controller, jog the wire inch keys to verify that Servo Torch is configured properly.

20. Multi-Equipment configurations change the equipment and repeat steps 7 -18. The Servo Torch axis index = 2 for the second servo drive.

Procedure 5-2 Setting Up Torch Guard

Components

- Torch Guard option

1. Load Torch Guard.
2. Set payload parameters as outlined. You will see a screen similar to the following.

```
MOTION PAYLOAD SET
Group 1
1 Schedule No[ 1]:[SERVO TORCH   ]
2 PAYLOAD       [kg]      5.00
3 PAYLOAD CENTER X [cm]        4.98
4 PAYLOAD CENTER Y [cm]        0.19
5 PAYLOAD CENTER Z [cm]       13.06
6 PAYLOAD INERTIA X [kgfcms^2]  0.10
7 PAYLOAD INERTIA Y [kgfcms^2]  0.33
8 PAYLOAD INERTIA Z [kgfcms^2]  0.00
```

3. Set active payload ID. You will see a screen similar to the following.

```
MOTION PERFORMANCE
  Group1              1/10
No.     PAYLOAD[kg]     Comment
 1        5.00 [SVT Binzel   ]
 2       20.00 [             ]
 3       20.00 [             ]
 4       20.00 [             ]
 5       20.00 [             ]
 6       20.00 [             ]
 7       20.00 [             ]
 8       20.00 [             ]
 9       20.00 [             ]
10       20.00 [             ]
```

4. Enable collision guard. You will see a screen similar to the following.

```
COL GUARD SETUP        JGFRM 100%
1 Collision Guard status: ENABLED
2 Sensitivity:    135%
3 Sensitivity Macro Reg:  R[15]
```

5. Set sensitivity.

Note Tune sensitivity per application.

5.2 Payload Setting

Setting Torch Payload and Activating Collision Guard for a FANUC Servo Torch with Binzel VTS-500

Payload ID Settings

Insert payload ID settings into the first schedule. Set the active payload to equal the correct schedule. See the following screen for an example.

```
  MOTION PAYLOAD\SET
 Group 1
1 Schedule No[   1]:[SERVO TORCH     ]
2 PAYLOAD         [kg]          5.00
3 PAYLOAD CENTER X  [cm]          4.98
4 PAYLOAD CENTER Y  [cm]          0.19
5 PAYLOAD CENTER Z  [cm]         13.06
6 PAYLOAD INERTIA X [kgfcms^2]    0.10
7 PAYLOAD INERTIA Y [kgfcms^2]    0.33
8 PAYLOAD INERTIA Z [kgfcms^2]    0.00
```

Active Payload

Set the active payload to equal the correct schedule. See the following screen for an example.

```
MOTION PERFORMANCE
Group1                         1/10    N
o.    PAYLOAD[kg]         Comment
\\1            5.00   [SVT Binzel      ]
2           20.00   [                ]
3           20.00   [                ]
4           20.00   [                ]
5           20.00   [                ]
6           20.00   [                ]
7           20.00   [                ]
8           20.00   [                ]
9           20.00   [                ]
10           20.00   [                ]
Active PAYLOAD number = 1
```

Collision Guard Activation

Enable Collision Guard and set the sensitivity to the application. Utilize the sensitivity command within the .TP program to fine tune performance dynamically. See the following screen for an example.

```
COL GUARD SETUP               JGFRM 100 %
1 Collision Guard status:     ENABLED
2 Sensitivity:                135%   (tune per application)
3 Sensitivity Macro Reg.:   R[ 15]
```

Glossary

A

abort

Abnormal termination of a computer program caused by hardware or software malfunction or operator cancellation.

absolute pulse code system

A positional information system for servomotors that relies on battery-backed RAM to store encoder pulse counts when the robot is turned off. This system is calibrated when it is turned on.

A/D value

An analog to digital-value. Converts a multilevel analog electrical system pattern into a digital bit.

AI

Analog input.

AO

Analog output.

alarm

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

algorithm

A fixed step-by-step procedure for accomplishing a given result.

alphanumeric

Data that are both alphabetical and numeric.

AMPS

Amperage amount.

analog

The representation of numerical quantities by measurable quantities such as length, voltage or resistance. Also refers to analog type I/O blocks and distinguishes them from discrete I/O blocks. Numerical data that can vary continuously, for example, voltage levels that can vary within the range of -10 to +10 volts.

AND

An operation that places two contacts or groups of contacts in series. All contacts in series control the resulting status and also mathematical operator.

ANSI

American National Standard Institute, the U.S. government organization with responsibility for the development and announcement of technical data standards.

APC

See absolute pulse code system.

APC motor

See servomotor.

application program

The set of instructions that defines the specific intended tasks of robots and robot systems to make them reprogrammable and multifunctional. You can initiate and change these programs.

arm

A robot component consisting of an interconnecting set of links and powered joints that move and support the wrist socket and end effector.

articulated arm

A robot arm constructed to simulate the human arm, consisting of a series of rotary motions and joints, each powered by a motor.

ASCII

Abbreviation for American Standard Code for Information Interchange. An 8-level code (7 bits plus 1 parity bit) commonly used for the exchange of data.

automatic mode

The robot state in which automatic operation can be initiated.

automatic operation

The time during which robots are performing programmed tasks through unattended program execution.

axis

1. A straight line about which a robot joint rotates or moves. 2. One of the reference lines or a coordinate system. 3. A single joint on the robot arm.

B

backplane

A group of connectors mounted at the back of a controller rack to which printed circuit boards are mated.

BAR

A unit of pressure equal to 100,000 pascals.

barrier

A means of physically separating persons from the restricted work envelope; any physical boundary to a hazard or electrical device/component.

battery low alarm

A programmable value (in engineering units) against which the analog input signal automatically is compared on Genius I/O blocks. A fault is indicated if the input value is equal to or less than the low alarm value.

baud

A unit of transmission speed equal to the number of code elements (bits) per second.

big-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

binary

A numbering system that uses only 0 and 1.

bit

Contraction of binary digit. 1. The smallest unit of information in the binary numbering system, represented by a 0 or 1. 2. The smallest division of a programmable controller word.

bps

Bits per second.

buffer

A storage area in the computer where data is held temporarily until the computer can process it.

bus

A channel along which data can be sent.

bus controller
A Genius bus interface board for a programmable controller.

bus scan
One complete communications cycle on the serial bus.

Bus Switching Module
A device that switches a block cluster to one bus or the other of a dual bus.

byte
A sequence of binary digits that can be used to store a value from 0 to 255 and usually operated upon as a unit. Consists of eight bits used to store two numeric or one alpha character.

C

calibration
The process whereby the joint angle of each axis is calculated from a known reference point.

Cartesian coordinate system
A coordinate system whose axes (x, y, and z) are three intersecting perpendicular straight lines. The origin is the intersection of the axes.

Cartesian coordinates
A set of three numbers that defines the location of a point within a rectilinear coordinate system and consisting of three perpendicular axes (x, y, z).

cathode ray tube
A device, like a television set, for displaying information.

central processing unit
The main computer component that is made up of a control section and an arithmetic-logic section. The other basic units of a computer system are input/output units and primary storage.

channel
The device along which data flow between the input/output units of a computer and primary storage.

character
One of a set of elements that can be arranged in ordered groups to express information. Each character has two forms: 1. a man-intelligible form, the graphic, including the decimal digits 0-9, the letters A-Z, punctuation marks, and other formatting and control symbols; 2. a computer intelligible form, the code, consisting of a group of binary digits (bits).

circular
A MOTYPE option in which the robot tool center point moves in an arc defined by three points. These points can be positions or path nodes.

clear

To replace information in a storage unit by zero (or blank, in some machines).

closed loop

A control system that uses feedback. An open loop control system does not use feedback.

C-MOS RAM

Complementary metal-oxide semiconductor random-access memory. A read/write memory in which the basic memory cell is a pair of MOS (metal-oxide semiconductor) transistors. It is an implementation of S-RAM that has very low power consumption, but might be less dense than other S-RAM implementations.

coaxial cable

A transmission line in which one conductor is centered inside and insulated from an outer metal tube that serves as the second conductor. Also known as coax, coaxial line, coaxial transmission line, concentric cable, concentric line, concentric transmission line.

component

An inclusive term used to identify a raw material, ingredient, part or subassembly that goes into a higher level of assembly, compound or other item.

computer

A device capable of accepting information, applying prescribed processes to the information, and supplying the results of these processes.

configuration

The joint positions of a robot and turn number of wrist that describe the robot at a specified position. Configuration is designated by a STRING value and is included in positional data.

continuous path

A trajectory control system that enables the robot arm to move at a constant tip velocity through a series of predefined locations. A rounding effect of the path is required as the tip tries to pass through these locations.

continuous process control

The use of transducers (sensors) to monitor a process and make automatic changes in operations through the design of appropriate feedback control loops. While such devices historically have been mechanical or electromechanical, microcomputers and centralized control is now used, as well.

continuous production

A production system in which the productive equipment is organized and sequenced according to the steps involved to produce the product. Denotes that material flow is continuous during the production process. The routing of the jobs is fixed and set-ups are seldom changed.

controlled stop

A controlled stop controls robot deceleration until it stops. When a safety stop input such as a safety fence signal is opened, the robot decelerates in a controlled manner and then stops. After the robot stops, the Motor Control Contactor opens and drive power is removed.

controller

A hardware unit that contains the power supply, operator controls, control circuitry, and memory that directs the operation and motion of the robot and communications with external devices. See control unit.

controller memory

A medium in which data are retained. Primary storage refers to the internal area where the data and program instructions are stored for active use, as opposed to auxiliary or external storage (magnetic tape, disk, diskette, and so forth.)

control, open-loop

An operation where the computer applies control directly to the process without manual intervention.

control unit

The portion of a computer that directs the automatic operation of the computer, interprets computer instructions, and initiates the proper signals to the other computer circuits to execute instructions.

coordinate system

See Cartesian coordinate system.

CPU

See central processing unit.

CRT

See cathode ray tube.

cps (viscosity)

Centipoises per second.

CRT/KB

Cathode ray tube/keyboard. An optional interface device for the robot system. The CRT/KB is used for some robot operations and for entering programs. It can be a remote device that attaches to the robot via a cable.

cycle

1. A sequence of operations that is repeated regularly. The time it takes for one such sequence to occur. 2. The interval of time during which a system or process, such as seasonal demand or a manufacturing operation, periodically returns to similar initial conditions. 3. The interval of time during which an event or set of events is completed. In production control, a cycle is the length of time between the release of a manufacturing order and shipment to the customer or inventory.

cycle time

1. In industrial engineering, the time between completion of two discrete units of production. 2. In materials management, the length of time from when material enters a production facility until it exits. See throughput.

cursor

An indicator on a teach pendant or CRT display screen at which command entry or editing occurs. The indicator can be a highlighted field or an arrow (> or ^).

cylindrical

Type of work envelope that has two linear major axes and one rotational major axis. Robotic device that has a predominantly cylindrical work envelope due to its design. Typically has fewer than 6 joints and typically has only 1 linear axis.

D

D/A converter

A digital-to-analog converter. A device that transforms digital data into analog data.

D/A value

A digital-to-analog value. Converts a digital bit pattern into a multilevel analog electrical system.

daisy chain

A means of connecting devices (readers, printers, etc.) to a central processor by party-line input/output buses that join these devices by male and female connectors. The last female connector is shorted by a suitable line termination.

daisy chain configuration

A communications link formed by daisy chain connection of twisted pair wire.

data

A collection of facts, numeric and alphabetical characters, or any representation of information that is suitable for communication and processing.

data base

A data file philosophy designed to establish the independence of computer program from data files. Redundancy is minimized and data elements can be added to, or deleted from, the file designs without changing the existing computer programs.

DC

Abbreviation for direct current.

DEADMAN switch

A control switch on the teach pendant that is used to enable servo power. Pressing the DEADMAN switch while the teach pendant is on activates servo power and releases the robot brakes; releasing the switch deactivates servo power and applies the robot brakes.

debugging

The process of detecting, locating and removing mistakes from a computer program, or manufacturing control system. See diagnostic routine.

deceleration tolerance

The specification of the percentage of deceleration that must be completed before a motion is considered finished and another motion can begin.

default

The value, display, function or program automatically selected if you have not specified a choice.

deviation

Usually, the absolute difference between a number and the mean of a set of numbers, or between a forecast value and the actual data.

device

Any type of control hardware, such as an emergency-stop button, selector switch, control pendant, relay, solenoid valve, or sensor.

diagnostic routine

A test program used to detect and identify hardware/software malfunctions in the controller and its associated I/O equipment. See debugging.

diagnostics

Information that permits the identification and evaluation of robot and peripheral device conditions.

digital

A description of any data that is expressed in numerical format. Also, having the states On and Off only.

digital control

The use of a digital computer to perform processing and control tasks in a manner that is more accurate and less expensive than an analog control system.

digital signal

A single point control signal sent to or from the controller. The signal represents one of two states: ON (TRUE, 1. or OFF (FALSE, 0).

directory

A listing of the files stored on a device.

discrete

Consisting of individual, distinct entities such as bits, characters, circuits, or circuit components. Also refers to ON/OFF type I/O blocks.

disk

A secondary memory device in which information is stored on a magnetically sensitive, rotating disk.

disk memory

A non-programmable, bulk-storage, random-access memory consisting of a magnetized coating on one or both sides of a rotating thin circular plate.

drive power

The energy source or sources for the robot servomotors that produce motion.

DRAM

Dynamic Random Access Memory. A read/write memory in which the basic memory cell is a capacitor. DRAM (or D-RAM) tends to have a higher density than SRAM (or S-RAM). Due to the support circuitry required, and power consumption needs, it is generally impractical to use. A battery can be used to retain the content upon loss of power.

E

edit

1. A software mode that allows creation or alteration of a program. 2. To modify the form or format of data, for example, to insert or delete characters.

emergency stop

The operation of a circuit using hardware-based components that overrides all other robot controls, removes drive power from the actuators, and causes all moving parts of to stop. The operator panel and teach pendant are each equipped with EMERGENCY STOP buttons.

enabling device

A manually operated device that, when continuously activated, permits motion. Releasing the device stops the motion of the robot and associated equipment that might present a hazard.

encoder

1. A device within the robot that sends the controller information about where the robot is. 2. A transducer used to convert position data into electrical signals. The robot system uses an incremental optical encoder to provide position feedback for each joint. Velocity data is computed from the encoder signals and used as an additional feedback signal to assure servo stability.

end effector

An accessory device or tool specifically designed for attachment to the robot wrist or tool mounting plate to enable the robot to perform its intended tasks. Examples include gripper, spot weld gun, arc weld gun, spray paint gun, etc.

end-of-arm tooling

Any of a number of tools, such as welding guns, torches, bells, paint spraying devices, attached to the faceplate of the robot wrist. Also called end effector or EOAT.

engineering units

Units of measure as applied to a process variable, for example, psi, Degrees F., etc.

envelope, maximum

The volume of space encompassing the maximum designed movements of all robot parts including the end effector, workpiece, and attachments.

EOAT

See end of arm tooling, tool.

EPROM

Erasable Programmable Read Only Memory. Semiconductor memory that can be erased and reprogrammed. A non-volatile storage memory.

error

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

error message

A numbered message, displayed on the CRT/KB and teach pendant, that indicates a system problem or warns of a potential problem.

Ethernet

A Local Area Network (LAN) bus-oriented, hardware technology that is used to connect computers, printers, terminal concentrators (servers), and many other devices together. It consists of a master cable and connection devices at each machine on the cable that allow the various devices to "talk" to each other. Software that can access the Ethernet and cooperate with machines connected to the cable is necessary. Ethernets come in varieties such as baseband and broadband and can run on different media, such as coax, twisted pair and fiber. Ethernet is a trademark of Xerox Corporation.

execute

To perform a specific operation, such as one that would be accomplished through processing one statement or command, a series of statements or commands, or a complete program or command procedure.

extended axis

An optional, servo-controlled axis that provides extended reach capability for a robot, including in-booth rail, single- or double-link arm, also used to control motion of positioning devices.

F

faceplate

The tool mounting plate of the robot.

feedback

1. The signal or data fed back to a commanding unit from a controlled machine or process to denote its response to the command signal. The signal representing the difference between actual response and desired response that is used by the commanding unit to improve performance of the controlled machine or process. 2. The flow of information back into the control system so that actual performance can be compared with planned performance, for instance in a servo system.

field

A specified area of a record used for a particular category of data. 2. A group of related items that occupy the same space on a CRT/KB screen or teach pendant LCD screen. Field name is the name of the field; field items are the members of the group.

field devices

User-supplied devices that provide information to the PLC (inputs: push buttons, limit switches, relay contacts, an so forth) or perform PLC tasks (outputs: motor starters, solenoids, indicator lights, and so forth.)

file

1. An organized collection of records that can be stored or retrieved by name. 2. The storage device on which these records are kept, such as bubble memory or disk.

filter

A device to suppress interference that would appear as noise.

Flash File Storage

A portion of FROM memory that functions as a separate storage device. Any file can be stored on the FROM disk.

Flash ROM

Flash Read Only Memory. Flash ROM is not battery-backed memory but it is non-volatile. All data in Flash ROM is saved even after you turn off and turn on the robot.

flow chart

A systems analysis tool to graphically show a procedure in which symbols are used to represent operations, data, flow, and equipment. See block diagram, process chart.

flow control

A specific production control system that is based primarily on setting production rates and feeding work into production to meet the planned rates, then following it through production to make sure that it is moving. This concept is most successful in repetitive production.

format

To set up or prepare a memory card or floppy disk (not supported with version 7.20 and later) so it can be used to store data in a specific system.

FR

See Flash ROM.

F-ROM

See Flash ROM.

FROM disk

See Flash ROM.

G

general override stat

A percentage value that governs the maximum robot jog speed and program run speed.

Genius I/O bus

The serial bus that provides communications between blocks, controllers, and other devices in the system especially with respect to GE FANUC Genius I/O.

gripper

The "hand" of a robot that picks up, holds and releases the part or object being handled. Sometimes referred to as a manipulator. See EOAT, tool.

group signal

An input/output signal that has a variable number of digital signals, recognized and taken as a group.

gun

See applicator.

H

Hand Model.

Used in Interference Checking, the Hand Model is the set of virtual model elements (spheres and cylinders) that are used to represent the location and shape of the end of arm tooling with respect to the robot's faceplate.

hardware

1. In data processing, the mechanical, magnetic, electrical and electronic devices of which a computer, controller, robot, or panel is built. 2. In manufacturing, relatively standard items such as nuts, bolts, washers, clips, and so forth.

hard-wire

To connect electric components with solid metallic wires.

hard-wired

1. Having a fixed wired program or control system built in by the manufacturer and not subject to change by programming. 2. Interconnection of electrical and electronic devices directly through physical wiring.

hazardous motion

Unintended or unexpected robot motion that can cause injury.

hexadecimal

A numbering system having 16 as the base and represented by the digits 0 through 9, and A through F.

hold

A smoothly decelerated stopping of all robot movement and a pause of program execution. Power is maintained on the robot and program execution generally can be resumed from a hold.

HTML.

Hypertext Markup Language. A markup language that is used to create hypertext and hypermedia documents incorporating text, graphics, sound, video, and hyperlinks.

http.

Hypertext transfer protocol. The protocol used to transfer HTML files between web servers.

I

impedance

A measure of the total opposition to current flow in an electrical circuit.

incremental encoder system

A positional information system for servomotors that requires calibrating the robot by moving it to a known reference position (indicated by limit switches) each time the robot is turned on or calibration is lost due to an error condition.

index

An integer used to specify the location of information within a table or program.

index register

A memory device containing an index.

industrial robot

A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions in order to perform a variety of tasks.

industrial robot system

A system that includes industrial robots, end effectors, any equipment devices and sensors required for the robot to perform its tasks, as well as communication interfaces for interlocking, sequencing, or monitoring the robot.

information

The meaning derived from data that have been arranged and displayed in a way that they relate to that which is already known. See data.

initialize

1. Setting all variable areas of a computer program or routine to their desired initial status, generally done the first time the code is executed during each run. 2. A program or hardware circuit that returns a program a system, or hardware device to an original state. See startup, initial.

input

The data supplied from an external device to a computer for processing. The device used to accomplish this transfer of data.

input device

A device such as a terminal keyboard that, through mechanical or electrical action, converts data from the form in which it has been received into electronic signals that can be interpreted by the CPU or programmable controller. Examples are limit switches, push buttons, pressure switches, digital encoders, and analog devices.

input processing time

The time required for input data to reach the microprocessor.

input/output

Information or signals transferred between devices, discreet electrical signals for external control.

input/output control

A technique for controlling capacity where the actual output from a work center is compared with the planned output developed by CRP. The input is also monitored to see if it corresponds with plans so that work centers will not be expected to generate output when jobs are not available to work on.

integrated circuit

A solid-state micro-circuit contained entirely within a chip of semiconductor material, generally silicon. Also called chip.

interactive

Refers to applications where you communicate with a computer program via a terminal by entering data and receiving responses from the computer.

interface

1. A concept that involves the specifications of the inter-connection between two equipments having different functions. 2. To connects a PLC with the application device, communications channel, and peripherals through various modules and cables. 3. The method or equipment used to communicate between devices.

interference zone

An area that falls within the work envelope of a robot, in which there is the potential for the robot motion to coincide with the motion of another robot or machine, and for a collision to occur.

interlock

An arrangement whereby the operation of one control or mechanism brings about, or prevents, the operations of another.

interrupt

A break in the normal flow of a system or program that occurs in a way that the flow can be resumed from that point at a later time. Interrupts are initiated by two types of signals: 1. signals originating within the computer system to synchronize the operation of the computer system with the outside world; 2. signals originating exterior to the computer system to synchronize the operation of the computer system with the outside world.

I/O

Abbreviation for input/output or input/output control.

I/O block

A microprocessor-based, configurable, rugged solid state device to which field I/O devices are attached.

I/O electrical isolation

A method of separating field wiring from logic level circuitry. This is typically done through optical isolation devices.

I/O module

A printed circuit assembly that is the interface between user devices and the Series Six PLC.

I/O scan

A method by which the CPU monitors all inputs and controls all outputs within a prescribed time. A period during which each device on the bus is given a turn to send information and listen to all of the broadcast data on the bus.

ISO

The International Standards Organization that establishes the ISO interface standards.

isolation

1. The ability of a logic circuit having more than one inputs to ensure that each input signal is not affected by any of the others. 2. A method of separating field wiring circuitry from logic level circuitry, typically done optically.

item

1. A category displayed on the teach pendant on a menu. 2. A set of adjacent digits, bits, or characters that is treated as a unit and conveys a single unit of information. 3. Any unique manufactured or purchased part or assembly: end product, assembly, subassembly, component, or raw material.

J

jog coordinate systems

Coordinate systems that help you to move the robot more effectively for a specific application. These systems include JOINT, WORLD, TOOL, and USER.

JOG FRAME

A jog coordinate system you define to make the robot jog the best way possible for a specific application. This can be different from world coordinate frame.

jogging

Pressing special keys on the teach pendant to move the robot.

jog speed

Is a percentage of the maximum speed at which you can jog the robot.

joint

1. A single axis of rotation. There are up to six joints in a robot arm (P-155 swing arm has 8). 2. A jog coordinate system in which one axis is moved at a time.

JOINT

A motion type in which the robot moves the appropriate combination of axes independently to reach a point most efficiently. (Point to point, non-linear motion).

joint interpolated motion

A method of coordinating the movement of the joints so all joints arrive at the desired location at the same time. This method of servo control produces a predictable path regardless of speed and results in the fastest cycle time for a particular move. Also called joint motion.

K

K

Abbreviation for kilo, or exactly 1024 in computer jargon. Related to 1024 words of memory.

KAREL

The programming language developed for robots by the FANUC America Corporation.

L

label

An ordered set of characters used to symbolically identify an instruction, a program, a quantity, or a data area.

LCD

See liquid crystal display.

lead time

The span of time needed to perform an activity. In the production and inventory control context, this activity is normally the procurement of materials and/or products either from an outside supplier or from one's own manufacturing facility. Components of lead time can include order preparation time, queue time, move or transportation time, receiving and inspection time.

LED

See Light Emitting Diode.

LED display

An alphanumeric display that consists of an array of LEDs.

Light Emitting Diode

A solid-state device that lights to indicate a signal on electronic equipment.

limiting device

A device that restricts the work envelope by stopping or causing to stop all robot motion and that is independent of the control program and the application programs.

limit switch

A switch that is actuated by some part or motion of a machine or equipment to alter the electrical circuit associated with it. It can be used for position detection.

linear

A motion type in which the appropriate combination of axes move in order to move the robot TCP in a straight line while maintaining tool center point orientation.

liquid crystal display

A digital display on the teach pendant that consists of two sheets of glass separated by a sealed-in, normally transparent, liquid crystal material. Abbreviated LCD.

little-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

load

1. The weight (force) applied to the end of the robot arm. 2. A device intentionally placed in a circuit or connected to a machine or apparatus to absorb power and convert it into the desired useful form. 3. To copy programs or data into memory storage.

location

1. A storage position in memory uniquely specified by an address. 2. The coordinates of an object used in describing its x, y, and z position in a Cartesian coordinate system.

lockout/tagout

The placement of a lock and/or tag on the energy isolating device (power disconnecting device) in the off or open position. This indicates that the energy isolating device or the equipment being controlled will not be operated until the lock/tag is removed.

log

A record of values and/or action for a given function.

logic

A fixed set of responses (outputs) to various external conditions (inputs). Also referred to as the program.

loop

The repeated execution of a series of instructions for a fixed number of times, or until interrupted by the operator.

M

mA

See milliampere.

machine language

A language written in a series of bits that are understandable by, and therefore instruct, a computer. This is a "first level" computer language, as compared to a "second level" assembly language, or a "third level" compiler language.

machine lock

A test run option that allows the operator to run a program without having the robot move.

macro

A source language instruction from which many machine-language instructions can be generated.

magnetic disk

A metal or plastic floppy disk (not supported on version 7.10 and later) that looks like a phonograph record whose surface can store data in the form of magnetized spots.

magnetic disk storage

A storage device or system consisting of magnetically coated metal disks.

magnetic tape

Plastic tape, like that used in tape recorder, on which data is stored in the form of magnetized spots.

maintenance

Keeping the robots and system in their proper operating condition.

MC

See memory card.

mechanical unit

The robot arm, including auxiliary axis, and hood/deck and door openers.

medium

plural **media** . The physical substance upon which data is recorded, such as a memory card (or floppy disk which is not supported on version 7.10 and later).

memory

A device or media used to store information in a form that can be retrieved and is understood by the computer or controller hardware. Memory on the controller includes C-MOS RAM, Flash ROM and D-RAM.

memory card

A C-MOS RAM memory card or a flash disk-based PC card.

menu

A list of options displayed on the teach pendant screen.

message

A group of words, variable in length, transporting an item of information.

microprocessor

A single integrated circuit that contains the arithmetic, logic, register, control and memory elements of a computer.

microsecond

One millionth (0.000001) of a second

milliampere

One one-thousandth of an ampere. Abbreviated mA.

millisecond

One thousandth of a second. Abbreviated msec.

module

A distinct and identifiable unit of computer program for such purposes as compiling, loading, and linkage editing. It is eventually combined with other units to form a complete program.

motion type

A feature that allows you to select how you want the robot to move from one point to the next. MOTYPES include joint, linear, and circular.

mode

1. One of several alternative conditions or methods of operation of a device. 2. The most common or frequent value in a group of values.

N

network

1. The interconnection of a number of devices by data communication facilities. "Local networking" is the communications network internal to a robot."Global networking" is the ability to provide communications connections outside of the robot's internal system. 2. Connection of geographically separated computers and/or terminals over communications lines. The control of transmission is managed by a standard protocol.

non-volatile memory

Memory capable of retaining its stored information when power is turned off.

O

Obstacle Model.
Used in Interference Checking, the Obstacle Model is the set of virtual model elements (spheres, cylinders, and planes) that are used to represent the shape and the location of a given obstacle in space.

off-line
Equipment or devices that are not directly connected to a communications line.

off-line operations
Data processing operations that are handled outside of the regular computer program. For example, the computer might generate a report off-line while the computer was doing another job.

off-line programming
The development of programs on a computer system that is independent of the "on-board" control of the robot. The resulting programs can be copied into the robot controller memory.

offset
The count value output from a A/D converter resulting from a zero input analog voltage. Used to correct subsequent non-zero measurements also incremental position or frame adjustment value.

on-line
A term to describe equipment or devices that are connected to the communications line.

on-line processing
A data processing approach where transactions are entered into the computer directly, as they occur.

operating system
Lowest level system monitor program.

operating work envelope
The portion of the restricted work envelope that is actually used by the robot while it is performing its programmed motion. This includes the maximum the end-effector, the workpiece, and the robot itself.

operator
A person designated to start, monitor, and stop the intended productive operation of a robot or robot system.

operator box
A control panel that is separate from the robot and is designed as part of the robot system. It consists of the buttons, switches, and indicator lights needed to operate the system.

operator panel
A control panel designed as part of the robot system and consisting of the buttons, switches, and indicator lights needed to operate the system.

optional features
Additional capabilities available at a cost above the base price.

OR

An operation that places two contacts or groups of contacts in parallel. Any of the contacts can control the resultant status, also a mathematical operation.

orientation

The attitude of an object in space. Commonly described by three angles: rotation about x (w), rotation about y (p), and rotation about z (r).

origin

The point in a Cartesian coordinate system where axes intersect; the reference point that defines the location of a frame.

OT

See overtravel.

output

Information that is transferred from the CPU for control of external devices or processes.

output device

A device, such as starter motors, solenoids, that receive data from the programmable controller.

output module

An I/O module that converts logic levels within the CPU to a usable output signal for controlling a machine or process .

outputs

Signals, typically on or off, that controls external devices based upon commands from the CPU.

override

See general override.

overtravel

A condition that occurs when the motion of a robot axis exceeds its prescribed limits.

overwrite

To replace the contents of one file with the contents of another file when copying.

P

parity

The anticipated state, odd or even, of a set of binary digits.

parity bit

A binary digit added to an array of bits to make the sum of all bits always odd or always even.

parity check

A check that tests whether the number of ones (or zeros) in an array of binary digits is odd or even.

parity error
A condition that occurs when a computed parity check does not agree with the parity bit.

part
A material item that is used as a component and is not an assembly or subassembly.

pascal
A unit of pressure in the meter-kilogram-second system equivalent to one newton per square meter.

path
1. A variable type available in the KAREL system that consists of a list of positions. Each node includes positional information and associated data. 2. The trajectory followed by the TCP in a move.

PCB
See printed circuit board.

PC Interface
The PC Interface option provides the RPC functions and PC send macros required by applications created using PC Developer's Kit.

pendant
See teach pendant.

PLC
See programmable logic controller or cell controller.

PMC
The programmable machine controller (PMC) functions provide a ladder logic programming environment to create PMC functions. This provides the capability to use the robot I/O system to run PLC programs in the background of normal robot operations. This function can be used to control bulk supply systems, fixed automation that is part of the robot workcell, or other devices that would normally require basic PLC controls.

printed circuit board
A flat board whose front contains slots for integrated circuit chips and connections for a variety of electronic components, and whose back is printed with electrically conductive pathways between the components.

production mode
See automatic mode.

program
1. A plan for the solution of a problem. A complete program includes plans for the transcription of data, coding for the computer, and plans for the absorption of the results into the system. 2. A sequence of instructions to be executed by the computer or controller to control a robot/robot system. 3. To furnish a computer with a code of instructions. 4. To teach a robot system a specific set of movements and instructions to do a task.

programmable controller

See programmable logic controller or cell controller.

programmable logic controller

A solid-state industrial control device that receives inputs from user-supplied control devices, such as switches and sensors, implements them in a precise pattern determined by ladder diagram-based programs stored in the user memory, and provides outputs for control of processes or user-supplied devices such as relays and motor starters.

Program ToolBox

The Program ToolBox software provides programming utilities such as mirror image and flip wrist editing capabilities.

protocol

A set of hardware and software interfaces in a terminal or computer that allows it to transmit over a communications network, and that collectively forms a communications language.

psi

Pounds per square inch.

Q

queue.

1. Waiting lines resulting from temporary delays in providing service. 2. The amount of time a job waits at a work center before set-up or work is performed on the job. See also job queue.

R

RAM

See Random Access Memory.

random access

A term that describes files that do not have to be searched sequentially to find a particular record but can be addressed directly.

Random Access Memory

1. Volatile, solid-state memory used for storage of programs and locations; battery backup is required. 2. The working memory of the controller. Programs and variable data must be loaded into RAM before the program can execute or the data can be accessed by the program.

range

1. A characterization of a variable or function. All the values that a function can possess. 2. In statistics, the spread in a series of observations. 3. A programmable voltage or current spectrum of values to which input or output analog signals can be limited.

RI

Robot input.

RO

Robot output.

read

To copy, usually from one form of storage to another, particularly from external or secondary storage to internal storage. To sense the meaning of arrangements of hardware. To sense the presence of information on a recording medium.

Read Only Memory

A digital memory containing a fixed pattern of bits that you cannot alter.

record

To store the current set or sets of information on a storage device.

recovery

The restoration of normal processing after a hardware or software malfunction through detailed procedures for file backup, file restoration, and transaction logging.

register

1. A special section of primary storage in a computer where data is held while it is being worked on.
2. A memory device capable of containing one or more computer bits or words.

remote/local

A device connection to a given computer, with remote devices being attached over communications lines and local devices attached directly to a computer channel; in a network, the computer can be a remote device to the CPU controlling the network.

repair

To restore robots and robot systems to operating condition after damage, malfunction, or wear.

repeatability

The closeness of agreement among the number of consecutive movements made by the robot arm to a specific point.

reset

To return a register or storage location to zero or to a specified initial condition.

restricted work envelope

That portion of the work envelope to which a robot is restricted by limiting devices that establish limits that will not be exceeded in the event of any reasonably foreseeable failure of the robot or its controls. The maximum distance the robot can travel after the limited device is actuated defines the restricted work envelope of the robot.

RIA

Robotic Industries Association Subcommittee of the American National Standards Institute, Inc.

robot

A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices, through variable programmed motions for the performance of a variety of tasks.

Robot Model.

Used in Interference Checking, the Robot Model is the set of virtual model elements (sphere and cylinders) that are used to represent the location and shape of the robot arm with respect to the robot's base. Generally, the structure of a six axes robot can be accurately modeled as a series of cylinders and spheres. Each model element represents a link or part of the robot arm.

ROM

See Read Only Memory.

routine

1. A list of coded instructions in a program. 2. A series of computer instructions that performs a specific task and can be executed as often as needed during program execution.

S

saving data.

Storing program data in Flash ROM, to a floppy disk (not supported on version 7.10 and later), or memory card.

scfm

Standard cubic feet per minute.

scratch start

Allows you to enable and disable the automatic recovery function.

sensor

A device that responds to physical stimuli, such as heat, light, sound pressure, magnetism, or motion, and transmits the resulting signal or data for providing a measurement, operating a control or both. Also a device that is used to measure or adjust differences in voltage in order to control sophisticated machinery dynamically.

serial communication

A method of data transfer within a PLC whereby the bits are handled sequentially rather than simultaneously as in parallel transmission.

serial interface

A method of data transmission that permits transmitting a single bit at a time through a single line. Used where high speed input is not necessary.

Server Side Include (SSI)

A method of calling or "including" code into a web page.

servomotor
An electric motor that is controlled to produce precision motion. Also called a "smart" motor.

SI
System input.

signal
The event, phenomenon, or electrical quantity that conveys information from one point to another.

significant bit
A bit that contributes to the precision of a number. These are counted starting with the bit that contributes the most value, of "most significant bit", and ending with the bit that contributes the least value, or "least significant bit".

singulating
Separating parts into a single layer.

slip sheet
A sheet of material placed between certain layers of a unit load. Also known as tier sheet.

SO
System output.

specific gravity
The ratio of a mass of solid or liquid to the mass of an equal volume of water at 45C. You must know the specific gravity of the dispensing material to perform volume signal calibration. The specific gravity of a dispensing material is listed on the MSDS for that material.

SRAM
A read/write memory in which the basic memory cell is a transistor. SRAM (or S-RAM) tends to have a lower density than DRAM. A battery can be used to retain the content upon loss of power.

slpm
Standard liters per minute.

Standard Operator Panel (SOP).
A panel that is made up of buttons, keyswitches, and connector ports.

state
The on or off condition of current to and from and input or output device.

statement
See instruction.

storage device
Any device that can accept, retain, and read back one or more times. The available storage devices are SRAM, Flash ROM (FROM or F-ROM), floppy disks (not available on version 7.10 and later), memory cards, or a USB memory stick.

system variable

An element that stores data used by the controller to indicate such things as robot specifications, application requirements, and the current status of the system.

T

Tare

The difference between the gross weight of an object and its contents, and the object itself. The weight of an object without its contents.

TCP

See tool center point.

teaching

Generating and storing a series of positional data points effected by moving the robot arm through a path of intended motions.

teach mode

1. The mode of operation in which a robot is instructed in its motions, usually by guiding it through these motions using a teach pendant. 2. The generation and storage of positional data. Positional data can be taught using the teach pendant to move the robot through a series of positions and recording those positions for use by an application program.

teach pendant

1. A hand-held device used to instruct a robot, specifying the character and types of motions it is to undertake. Also known as teach box, teach gun. 2. A portable device, consisting of an LCD display and a keypad, that serves as a user interface to the KAREL system and attaches to the operator box or operator panel via a cable. The teach pendant is used for robot operations such as jogging the robot, teaching and recording positions, and testing and debugging programs.

telemetry

The method of transmission of measurements made by an instrument or a sensor to a remote location.

termination type

Feature that controls the blending of robot motion between segments.

tool

A term used loosely to define something mounted on the end of the robot arm, for example, a hand, gripper, or an arc welding torch.

tool center point

1. The location on the end-effector or tool of a robot hand whose position and orientation define the coordinates of the controlled object. 2. Reference point for position control, that is, the point on the tool that is used to teach positions. Abbreviated TCP.

TOOL Frame
The Cartesian coordinate system that has the position of the TCP as its origin to stet. The z-axis of the tool frame indicates the approach vector for the tool.

TP.
See teach pendant.

transducer
A device for converting energy from one form to another.

U

UOP
See user operator panel.

URL
Universal Resource Locator. A standard addressing scheme used to locate or reference files on web servers.

USB memory stick
The controller USB memory stick interface supports a USB 1.1 interface. The USB Organization specifies standards for USB 1.1 and 2.0. Most memory stick devices conform to the USB 2.0 specification for operation and electrical standards. USB 2.0 devices as defined by the USB Specification must be backward compatible with USB 1.1 devices. However, FANUC America Corporation does not support any security or encryption features on USB memory sticks. The controller supports most widely-available USB Flash memory sticks from 32MB up to 1GB in size.

USER Frame
The Cartesian coordinate system that you can define for a specific application. The default value of the User Frame is the World Frame. All positional data is recorded relative to User Frame.

User Operator Panel
User-supplied control device used in place of or in parallel with the operator panel or operator box supplied with the controller. Abbreviated UOP .

V

variable
A quantity that can assume any of a given set of values.

variance
The difference between the expected (or planned) and the actual, also statistics definitions.

vision system
A device that collects data and forms an image that can be interpreted by a robot computer to determine the position or to “see” an object.

volatile memory

Memory that will lose the information stored in it if power is removed from the memory circuit device.

W

web server

An application that allows you to access files on the robot using a standard web browser.

warning device

An audible or visible device used to alert personnel to potential safety hazards.

work envelope

The volume of space that encloses the maximum designed reach of the robot manipulator including the end effector, the workpiece, and the robot itself. The work envelope can be reduced or restricted by limiting devices. The maximum distance the robot can travel after the limit device is actuated is considered the basis for defining the restricted work envelope.

write

To deliver data to a medium such as storage.

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